

Recall: indeterminate form

$$\frac{\infty}{\infty}; \frac{0}{0}; 0 \cdot \infty,$$

$$\infty - \infty; 0^0; \infty^0; 1^\infty.$$

1. (Fall 2024, #1) Evaluate the limits.

$$(a) \lim_{x \rightarrow 0^+} (1 + x \ln(x))^{\frac{1}{x}}$$

Take $\ln$

$$\ln \left[(1 + x \ln x)^{\frac{1}{x}} \right]$$

$$= \frac{1}{x} \ln(1 + x \ln x)$$

$$\text{When } x \rightarrow 0^+ \quad \lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = 0 \quad (*)$$

$$\text{Then } \lim_{x \rightarrow 0^+} \ln(1 + x \ln x) = \ln(1) = 0.$$

$$\text{Then } \lim_{x \rightarrow 0^+} \frac{\ln(1 + x \ln x)}{x} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0^+} \frac{x \cdot \frac{1}{x} + \ln x}{1} = \lim_{x \rightarrow 0^+} \frac{1 + \ln x}{1 + x \ln x} \stackrel{\text{L'H}}{=} -\infty$$

$\Rightarrow$ The original limit is $e^{-\infty} = \boxed{0}$.

$$(b) \lim_{x \rightarrow 0} \frac{\sin x - x \cos x - \sin(x^3/3)}{x^5}$$

Rem:

$$\text{Note that } \lim_{x \rightarrow 0} \sin x - x \cos x - \sin(x^3/3) = 0,$$

(Here $\nearrow 0 \quad \nearrow 0 \quad \nearrow 0$
 $-x \leq x \cos x \leq x$ by squeeze rule).

$$\lim_{x \rightarrow 0^+} (1+x)^{\frac{1}{x}}$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

$$= e. \quad \stackrel{\text{L'H}}{\stackrel{\frac{0}{0}}{=}} \lim_{x \rightarrow 0} \frac{\cos x - (\cos x - x \sin x) - \cos(x^3/3) \cdot x^2}{5x^4}$$

$$= \lim_{x \rightarrow 0} \frac{x \sin x - x^2 \cos(x^3/3)}{5x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x - x \cos(x^3/3)}{5x^3} \quad \nearrow x^3 \sin(x^3/3).$$

$$\stackrel{\frac{0}{0}}{\stackrel{\text{L'H}}{=}} \lim_{x \rightarrow 0} \frac{\cos x - \cos(x^3/3) + x \sin(x^3/3) \cdot x^2}{15x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - \cos \frac{x^3}{3}}{15x^2} + \lim_{x \rightarrow 0} \frac{1}{15} \cdot x \sin \left(\frac{x^3}{3} \right)$$

$\hookrightarrow 0$

$$= \lim_{x \rightarrow 0} \frac{\cos x - \cos \frac{x^3}{3}}{15x^2}$$

"0/0" L.H

$$\stackrel{L.H}{=} \lim_{x \rightarrow 0} \frac{-\sin x + \sin \frac{x^3}{3} \cdot x^2}{30x}$$

$$= -\frac{1}{30} \lim_{x \rightarrow 0} \frac{\sin x}{x} + \lim_{x \rightarrow 0} \frac{1}{30} \cdot \sin \frac{x^3}{3} \cdot x$$

$\hookrightarrow 0$

$$= -\frac{1}{30} \quad \hookrightarrow 1$$

(One can also use Maclaurin expansion to find the limit. We will study this method in Topic C.)

2. (Spring 2024, #1) Evaluate the following limits.

$$(a) \lim_{x \rightarrow \infty} x \sinh\left(\frac{1}{x}\right) \quad \sinh(0) = 0$$

$$= \lim_{x \rightarrow \infty} \frac{\sinh\left(\frac{1}{x}\right)}{\frac{1}{x}}$$

$$\stackrel{u = \frac{1}{x}}{=} \lim_{u \rightarrow 0^+} \frac{\sinh(u)}{u}$$

$$\stackrel{0/0}{=} \lim_{u \rightarrow 0^+} \frac{\cosh(u)}{1} = 1.$$

$$(b) \lim_{x \rightarrow 0^+} x^{\sin x} \quad 0^0.$$

$$= \lim_{x \rightarrow 0^+} e^{\ln(x^{\sin x})}$$

$$= e^{\lim_{x \rightarrow 0^+} \sin x \ln x}$$

$$= e^{\lim_{x \rightarrow 0^+} \boxed{\frac{\sin x}{x}} \cdot x \ln x} \quad \nearrow 1.$$

$$= e^{\lim_{x \rightarrow 0^+} x \ln x}$$

$$= e^{\lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}}}$$

$$\stackrel{L'H \cdot \frac{0}{0}}{=} e^{\lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}} = e^0 = 1.$$

3. (Fall 2023, #1) Evaluate the following limits.

(a) $\lim_{x \rightarrow 0^+} (\tan^{-1}(x))^{x^2}$

$$= \lim_{x \rightarrow 0^+} e^{\ln((\tan^{-1}(x))^{x^2})}$$

$$= \lim_{x \rightarrow 0^+} e^{x^2 \ln(\tan^{-1}(x))}$$

$$= e^{\lim_{x \rightarrow 0^+} x^2 \ln(\tan^{-1}(x))}$$

$$= e^{\lim_{x \rightarrow 0^+} \frac{\ln(\tan^{-1}(x))}{\frac{1}{x^2}}}$$

"0/0"

$$\stackrel{\text{L'H}}{=} e^{\lim_{x \rightarrow 0^+} \frac{\frac{1}{\tan^{-1}(x)} \cdot \left(\frac{1}{1+x^2}\right)}{-2x^{-3}}}$$

→ 1

$$= e^{\lim_{x \rightarrow 0^+} \frac{x^3}{-2\tan^{-1}(x) \cdot (1+x^2)}} \stackrel{\text{L'H}}{=} e^{\lim_{x \rightarrow 0^+} \frac{3x^2}{-2 \cdot \frac{1}{1+x^2}}} = e^0 = 1$$

"0/0"

(b) $\lim_{x \rightarrow +\infty} \left(x \sin\left(\frac{1}{x}\right) - x \tan\left(\frac{1}{x}\right) \right)$

★ "∞ - ∞"

$$= \lim_{x \rightarrow +\infty} x \left(\sin\left(\frac{1}{x}\right) - \tan\left(\frac{1}{x}\right) \right)$$

$$= \lim_{x \rightarrow +\infty} \frac{\sin\left(\frac{1}{x}\right) - \tan\left(\frac{1}{x}\right)}{\frac{1}{x}}$$

$$\stackrel{u = \frac{1}{x}}{=} \lim_{u \rightarrow 0^+} \frac{\sin u - \tan u}{u}$$

$$\stackrel{\text{L'H}}{=} \lim_{u \rightarrow 0^+} \frac{\cos u - \frac{1}{\cos^2 u}}{1} = 1 - 1 = 0$$

"0/0"

Recall $(\tan u)' = \sec^2 u = \frac{1}{\cos^2 u}$

4. (Spring 2023, #1) Evaluate the following limits:

$$(a) \lim_{x \rightarrow 0} \frac{2 \cos x + x^2 - 2}{x^4}$$

$$\stackrel{0/0}{=} \text{L-H} \lim_{x \rightarrow 0} \frac{-2 \sin x + 2x}{4x^3}$$

$$\stackrel{0/0}{=} \text{L-H} \lim_{x \rightarrow 0^+} \frac{-2 \cos x + 2}{12x^2}$$

$$\stackrel{0/0}{=} \text{L-H} \lim_{x \rightarrow 0^+} \frac{2 \sin x}{24x}$$

$$= \frac{1}{12} \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = \frac{1}{12} \quad \text{or } \stackrel{0/0}{=} \lim_{x \rightarrow 0^+} \frac{\cos x}{1} = 1$$

$$= \frac{1}{12}$$

Rem: Can apply Maclaurin expansion as well.

$$(b) \lim_{x \rightarrow +\infty} (x^3 e^{\frac{2}{x^3}} - x^3)$$

$$\stackrel{\infty - \infty}{=}$$

$$= \lim_{x \rightarrow +\infty} x^3 (e^{\frac{2}{x^3}} - 1)$$

$$= \lim_{x \rightarrow +\infty} \frac{e^{\frac{2}{x^3}} - 1}{\frac{1}{x^3}}$$

$$\stackrel{u = \frac{1}{x^3}}{=} \lim_{u \rightarrow 0^+} \frac{e^{2u} - 1}{u}$$

$$\stackrel{0/0}{=} \text{L-H} \lim_{u \rightarrow 0^+} \frac{2e^{2u}}{1} = 2$$

5. (Fall 2022, #1) Evaluate the following limits:

$$(a) \lim_{x \rightarrow 1} \frac{\ln x - x + 1}{(x-1)^2}$$

$$\begin{array}{l} \text{"0/0"} \\ \text{"L-H"} \end{array} \lim_{x \rightarrow 1} \frac{\frac{1}{x} - 1}{2(x-1)}$$

$$\begin{array}{l} \text{"0/0"} \\ \text{"L-H"} \end{array} \lim_{x \rightarrow 1} \frac{-\frac{1}{x^2}}{2} = -\frac{1}{2}.$$

$$(b) \lim_{x \rightarrow +\infty} x^{\frac{\ln x}{x}}$$

$$= \lim_{x \rightarrow +\infty} e^{\ln(x^{\frac{\ln x}{x}})}$$

$$= e^{\lim_{x \rightarrow +\infty} \frac{\ln x}{x} \ln(x)}$$

$$\begin{array}{l} \text{"0/0"} \\ \text{"L-H"} \end{array} e^{\lim_{x \rightarrow +\infty} \frac{(\ln x)^2}{x}}$$

$$= e^{\lim_{x \rightarrow +\infty} 2 \cdot \frac{\ln x}{x}}$$

$$\begin{array}{l} \text{"0/0"} \\ \text{L-H} \end{array} e^{2 \lim_{x \rightarrow +\infty} \frac{\frac{1}{x}}{1}} = e^0 = 1.$$